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College of Osteopathic
Medicine

Histology of Blood

Foundations of Osteopathic Medicine

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.



Session Objectives

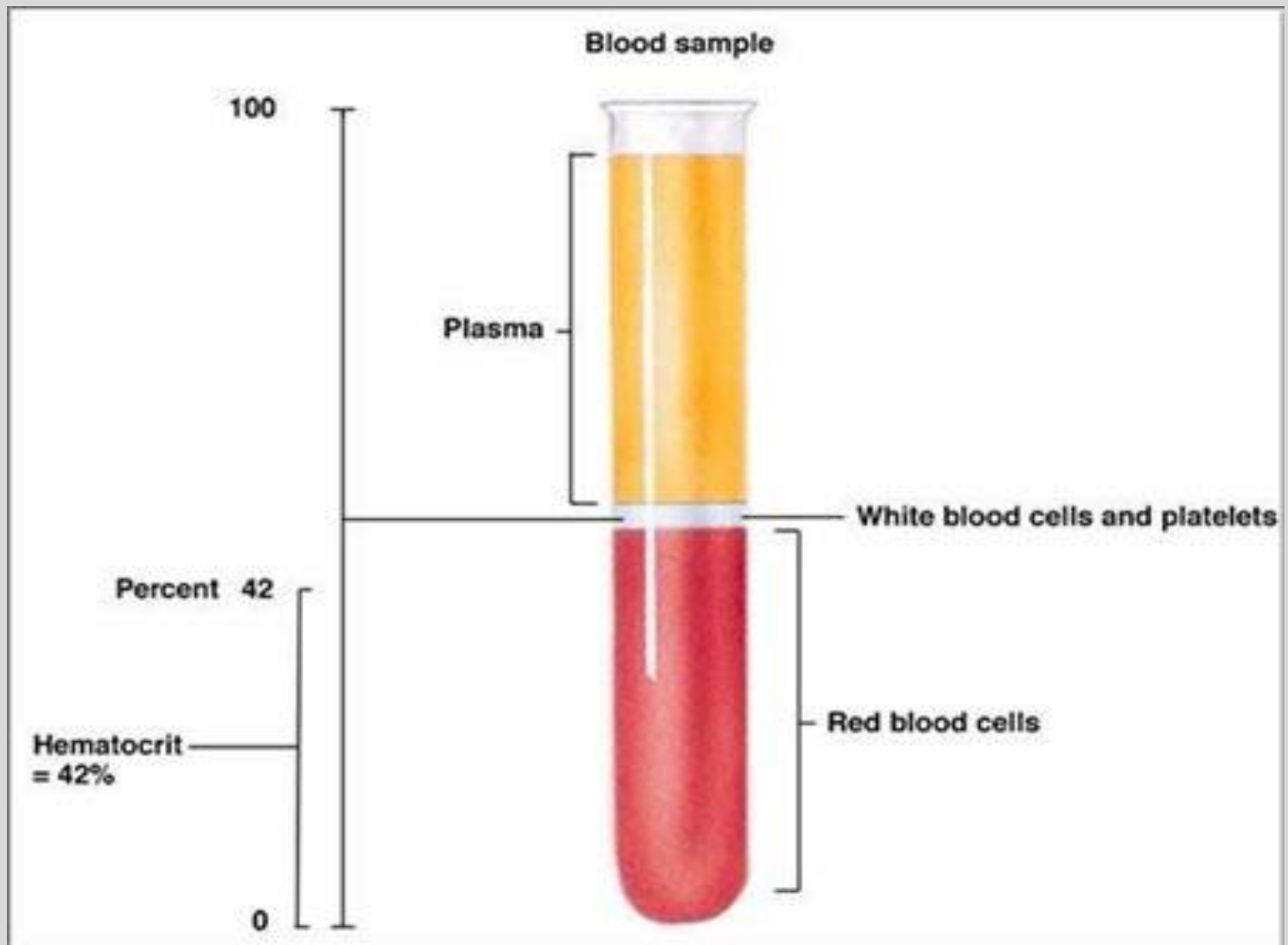
1. Describe and discuss the components of plasma, the concepts of hematocrit, and the general overview of CBC components.
2. Describe and discuss **erythrocytes** in terms of morphology, function, and their important role as a component of blood.
3. Describe, discuss, and compare **neutrophils, lymphocytes, and monocytes/macrophages** in terms of morphology, function, and their important roles as components of blood.
4. Describe, discuss, and compare **eosinophils and basophils** in terms of morphology, function, and their important roles as components of blood.
5. Describe and discuss **platelets** in terms of morphology and function and their important role as components of blood.

Topics

- Plasma
- Red blood Cells
- White Blood Cells
 - Granulocytes: neutrophils, eosinophils, basophils
 - Agranulocytes: lymphocytes, monocytes
- Platelets

INTRODUCTION

- Blood: connective tissue
- 5 – 6 L in human body
- Plasma, erythrocytes (rbcs), leukocytes (wbcs), and thrombocytes (platelets)
- Blood sample collected in test tube with anticoagulant (citrate, heparin, etc.) and centrifuged:
 - 45% plasma, 55% cells, 1% leukocytes and platelets (buffy coat)
- Hematocrit – volume of packed erythrocytes



Plasma

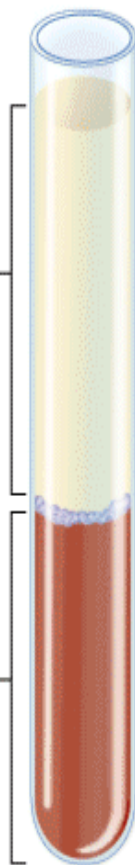
- 90% water, solvent
- Albumin – most abundant plasma protein, made in liver
 - Exerts concentration gradient between blood and ECF (colloid osmotic pressure)
 - carrier protein for hormones, metabolites, drugs (e.g. thyroxine, bilirubin, barbituates)
 - establishes gradient between blood and interstitial tissues

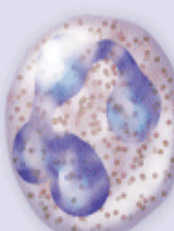
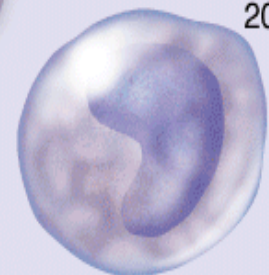
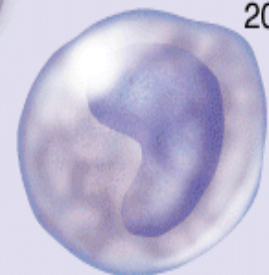
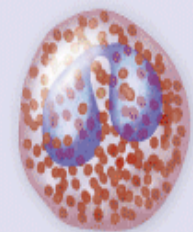
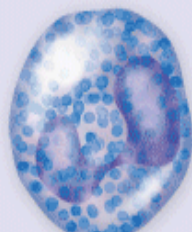
Plasma, continued

- Globulins – immunoglobulins and nonimmune globulins (latter also help maintain osmotic pressure and serve as carrier proteins)
 - nonimmune globulins include alpha and beta globulins, fibronectin, lipoproteins, coagulation factors
- Fibrinogen – important in clotting, made in liver
- Electrolytes, urea, creatinine, glucose, lipids, amino acids, oxygen, carbon dioxide, nitrogen, hormones, enzymes
- **SERUM – CLOTTING FACTORS HAVE BEEN REMOVED**

Plasma		
Water 92% by weight	Proteins 7% by weight	Other solutes 1% by weight
	Albumins 58%	Electrolytes
	Globulins 37%	Nutrients
	Fibrinogen 4%	Respiratory gases
	Regulatory proteins 1%	Waste products

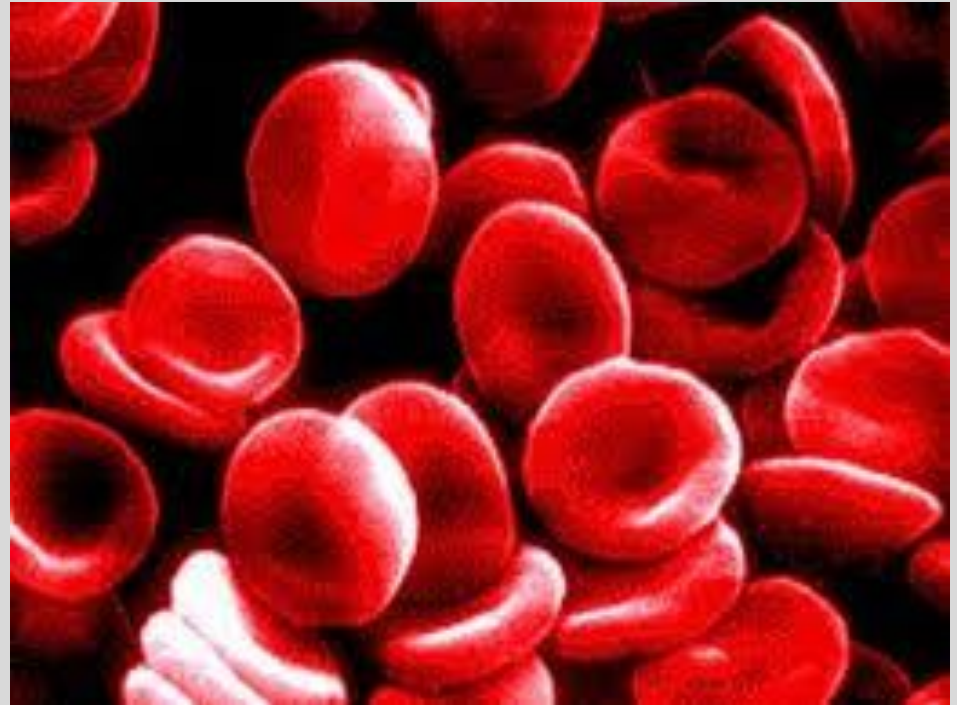
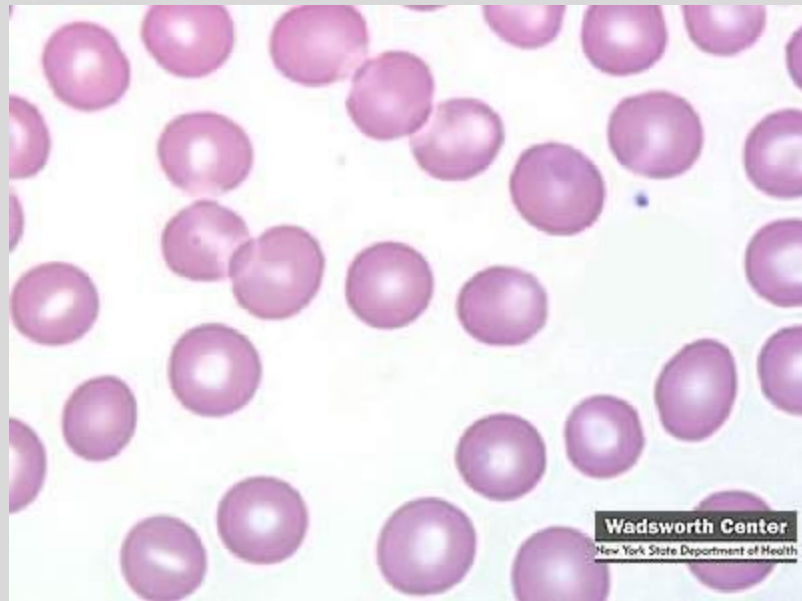
Erythrocytes
Erythrocytes 4.2–6.2 million per cubic mm

Buffy Coat	
Platelets 120–300 thousand per cubic mm 	Leukocytes 5–10 thousand per cubic mm  Neutrophils 60–70%  Lymphocytes 20–25%  Monocytes 3–8%  Eosinophils 2–4%  Basophils 0.5–1%

Source: Mescher AL: *Junqueira's Basic Histology: Text and Atlas*, 12th Edition: <http://www.accessmedicine.com>

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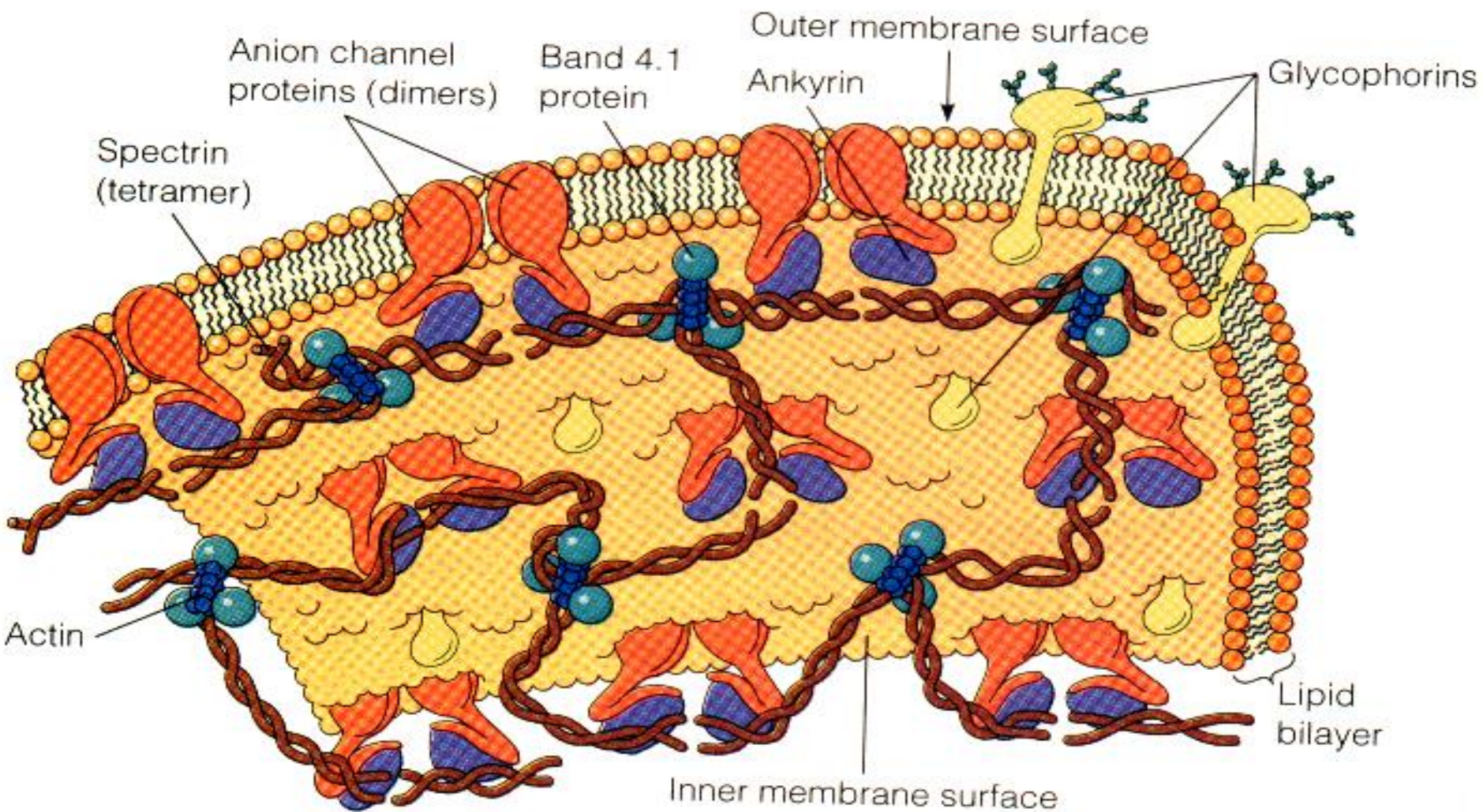
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Erythrocytes

- **Anucleate**
- Biconcave discs; shape maximizes surface area – important for gas exchange
- Bind oxygen for delivery to tissue
- Bind CO₂ for removal from tissue; 120 day lifespan
- Membrane is lipid bilayer –
 - contains integral membrane proteins (glycophorins and band 3 proteins)
 - peripheral membrane proteins (inner surface) which include spectrin tetramers, actin, tropomyosin, etc.

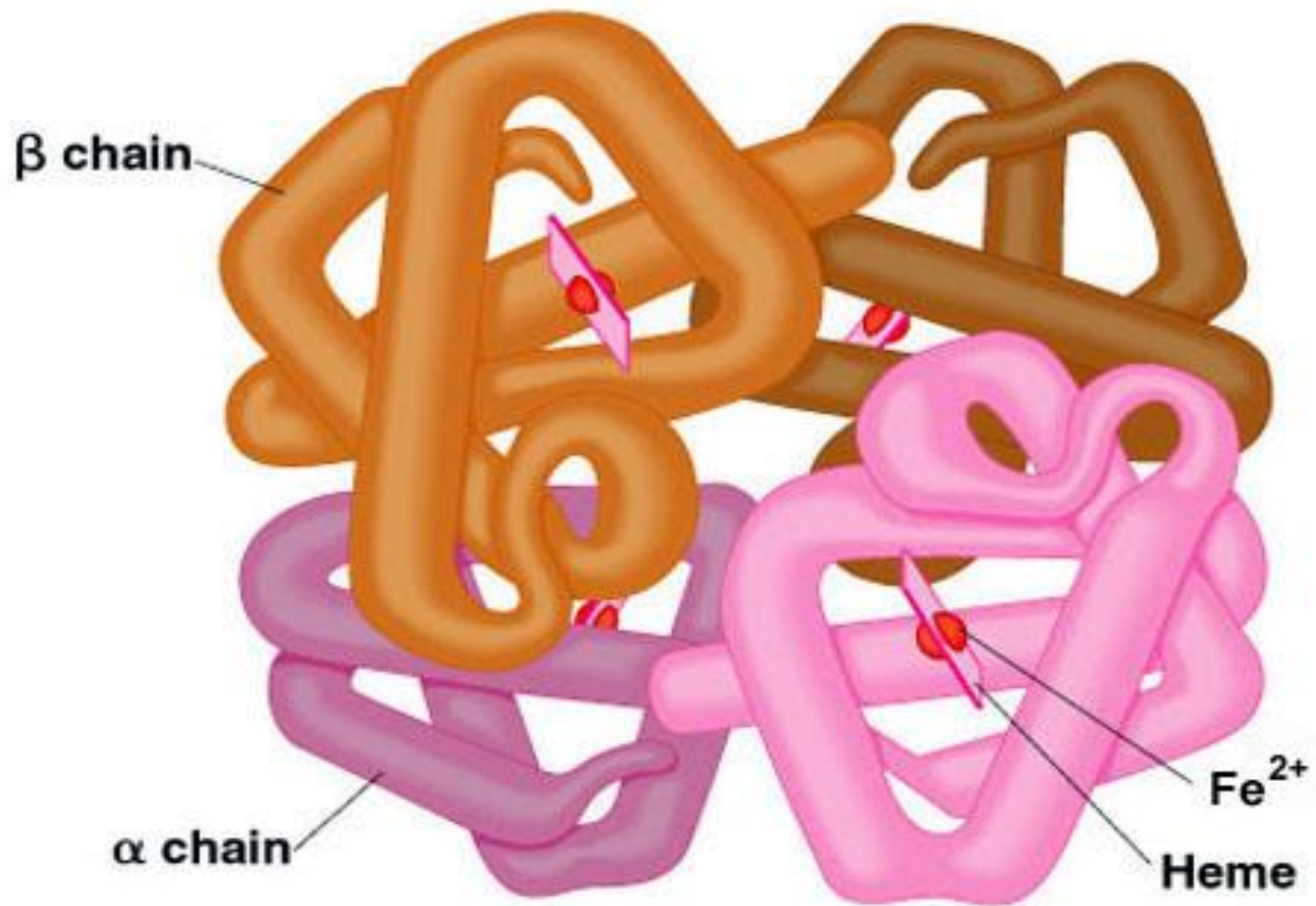
Erythrocytes

- Glycophorin C – important in attaching underlying cytoskeleton protein network to cell membrane
- Band 3 protein – binds hemoglobin, acts as additional anchoring site for cytoskeletal proteins
- Peripheral membrane proteins organized on inner surface into dimensional hexagonal lattice network, anchored to bilipid layer by **ankyrin**
- Unique structure contributes to shape of rbc and gives elastic properties and stability to membrane

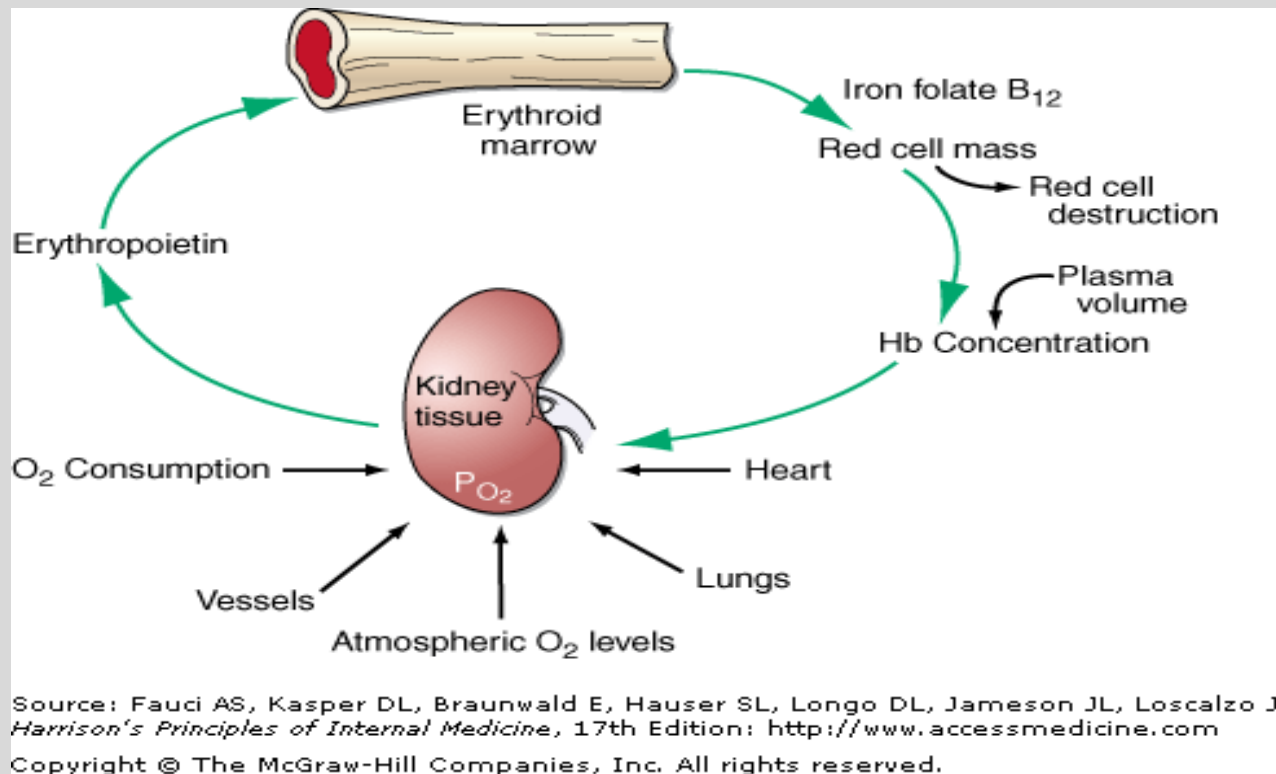


Erythrocytes

- Rbcs contain Hbg, protein which transports O₂ and CO₂.
 - Hbg has 4 polypeptide chains of alpha, beta, delta, and gamma each complexed to iron containing heme group
- **HbA** contains **2 alpha** and **2 beta chains** and is most prevalent in **adults**
- **HbA2** contains **2 alpha** and **2 delta chains** and represents about **2% of hemoglobin in adults**
- **HbF** contains **2 alpha** and **2 gamma** chains and is the major form in the **fetus**



RBC regulation



Red Blood Cells (RBC) Normal Lab Values

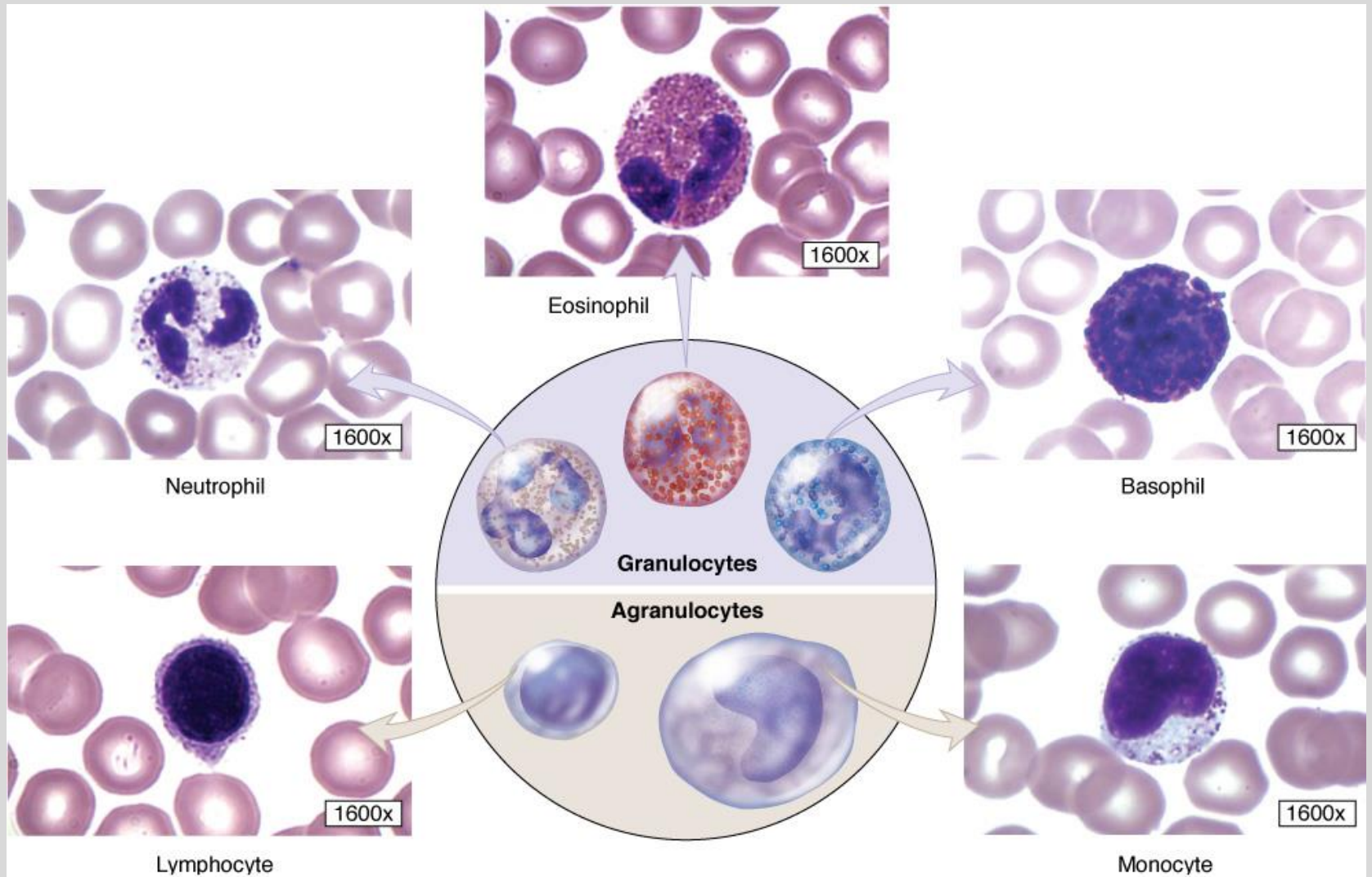
	Range (SI Units)	Range (Conventional)
Adult male	$4.7\text{—}6.2 \times 10^{12}/\text{L}$	$4.7\text{—}6.2 \times 10^6/\text{mm}^3$
Adult female	$4.5\text{—}5.0 \times 10^{12}/\text{L}$	$4.5\text{—}5.0 \times 10^6/\text{mm}^3$
Child	$3.5\text{—}5.5 \times 10^{12}/\text{L}$	$3.5\text{—}5.5 \times 10^6/\text{mm}^3$
<u>Newborn</u>	$4.8\text{—}7.1 \times 10^{12}/\text{L}$	$4.8\text{—}7.1 \times 10^6/\text{mm}^3$

Normal Lab Values for RBC Indices

	Range (SI Units)	Range (Conventional)
Mean corpuscular volume (MCV)	82—98 fL	82—98 mm ³
Mean corpuscular hemoglobin (MCH)	0.40—0.53 fmol/cell	26—34 pg/cel
Mean corpuscular hemoglobin concentration (MCHC)	320—360 g/L	32—36 g/dL
Reticulocyte Count	0.005—0.015	0.5%—1.5% of total RBCs

White Blood Cells

- **GRANULOCYTES:**
- **Neutrophils**
 - lobulated nucleus **40-70%** of circulating leukocytes
 - Life span 6-7 hours, may live up to 4 days in connective tissue
 - granules
 - bacterial infections and acute inflammation
- **Eosinophils**
 - bilobed nucleus, eosinophilic granules
 - **0-8%** of circulating leukocytes
 - parasites and bronchial asthma, allergies
- **Basophils**
 - basophilic
 - Less than **0-3%** circulating leukocytes
 - allergic reactions
- **AGRANULOCYTES**
 - **Lymphocytes:** chronic, viral, T lymphocytes, B lymphocytes, NK lymphocytes
 - **Monocytes:** become macrophages in tissue - phagocytosis



Source: Mescher AL: *Junqueira's Basic Histology: Text and Atlas*, 12th Edition: <http://www.accessmedicine.com>
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TABLE 41.2 Differential White Blood Cell Count

Cell Type	Normal Value (percent)	Elevated Levels May Indicate
Neutrophil	54–62	Bacterial infections, stress
Lymphocyte	25–33	Mononucleosis, whooping cough, viral infections
Monocyte	3–9	Malaria, tuberculosis, fungal infections
Eosinophil	1–3	Allergic reactions, autoimmune diseases, parasitic worms
Basophil	<1	Cancers, chicken pox, hypothyroidism

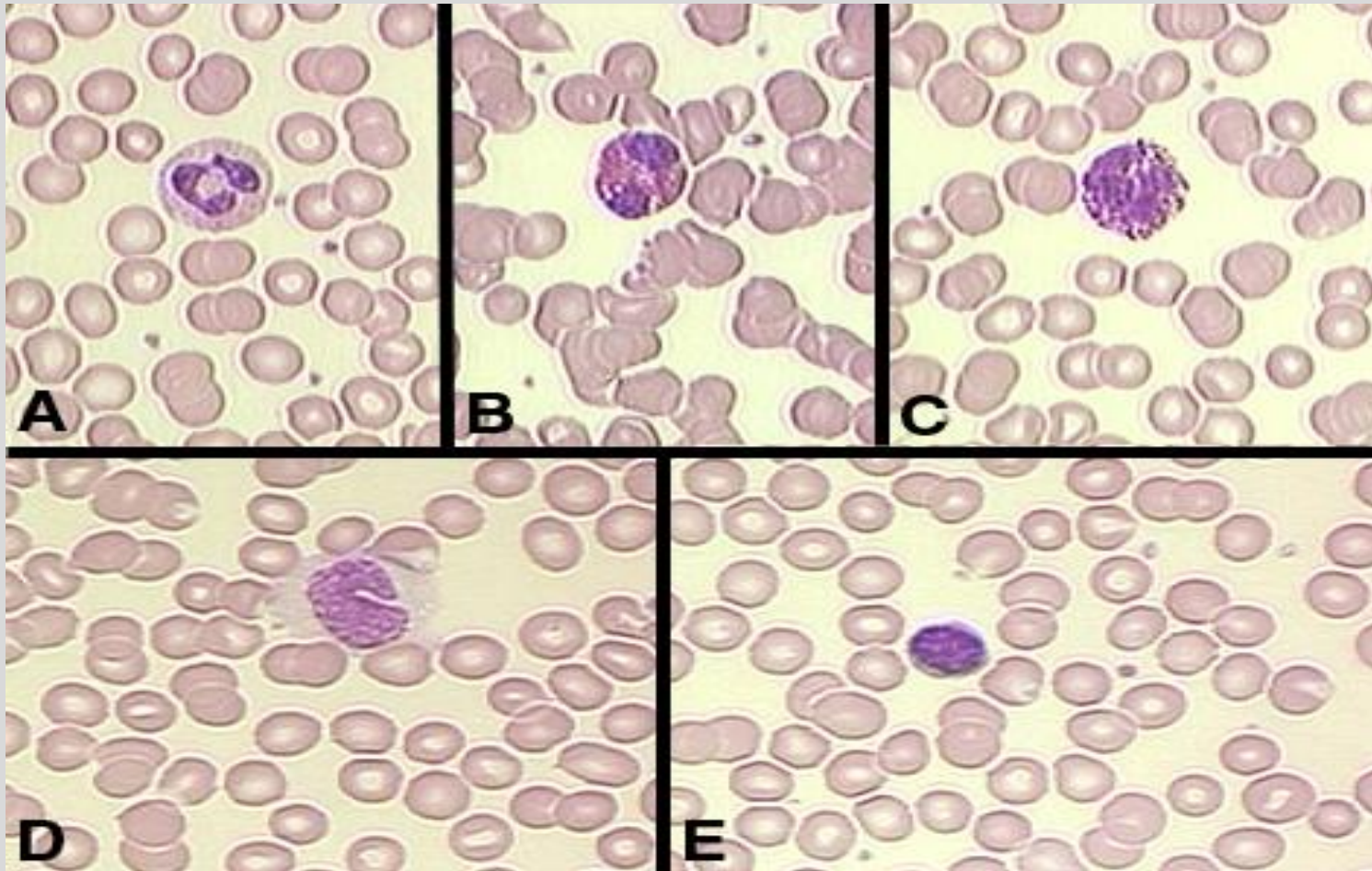
<https://www.chegg.com/homework-help/questions-and-answers/results-differential-white-blood-cell-count-compare-normal-values-listed-table-412-q29928149>

Normal Differential Count

Cell Type	%
Neutrophils	50-70
Eosinophils	1-3
Basophils	0-1
Lymphocytes	25-35
Monocytes	4-6

<https://www.getbodysmart.com/circulatory-system/white-blood-cell-differential>

CBC



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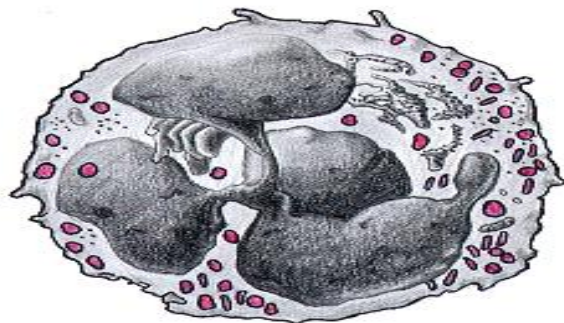
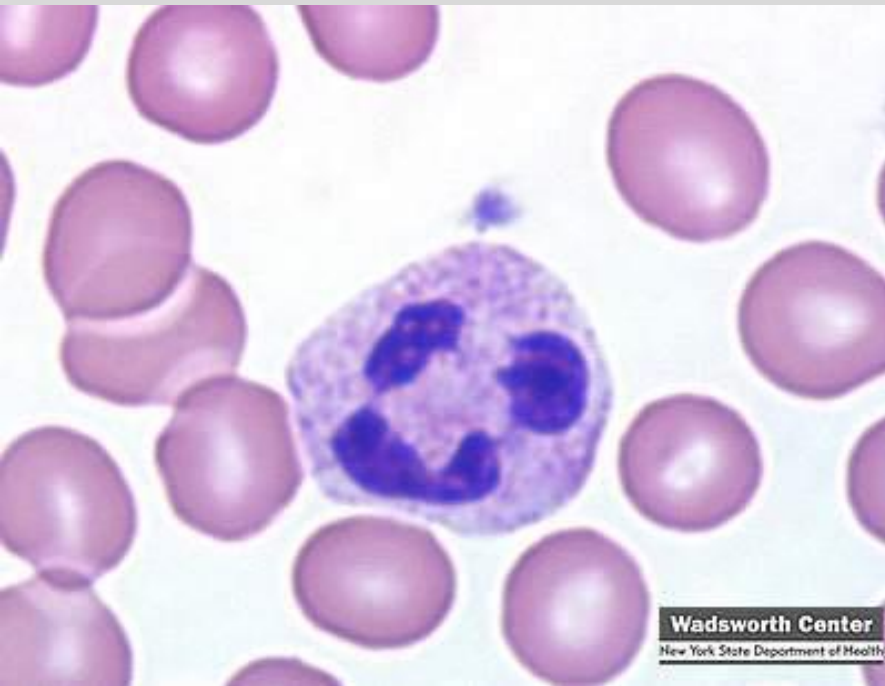


Fig. 8 - Neutrophil

NEUTROPHILS

- 3 types of granules
- motile cell
- active phagocytes at inflammation site

Neutrophils-Think Acute

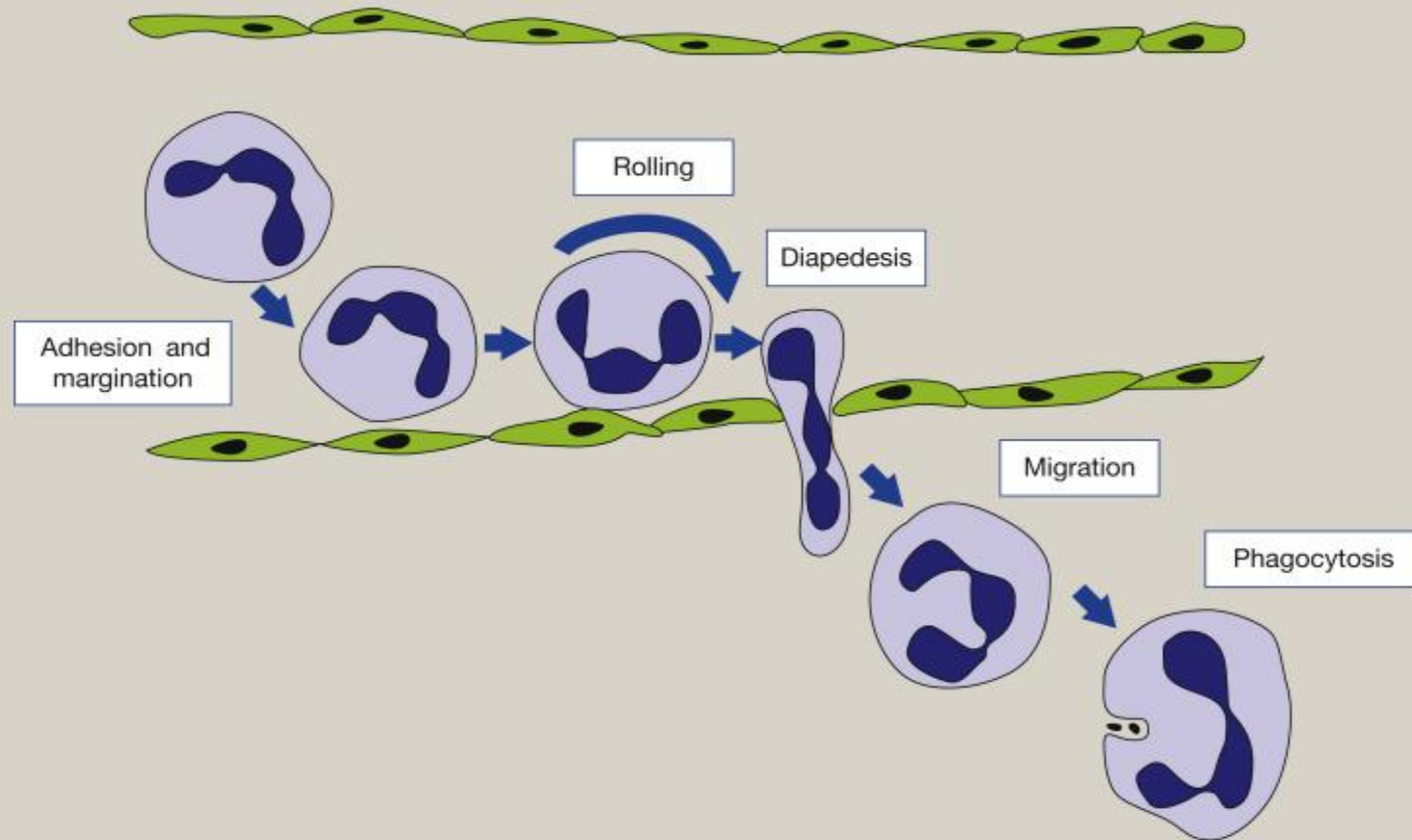
- **First responder and most numerous wbc, esp. present in bacterial infections**
- **MULTI-LOBED nuclei (2-4 lobes)**
- **THREE types of granules:**
 - **Azuropilic granules = primary granules →**
 - Found in all wbcs
 - less in # compared to specific granules
 - **LYSOSOMES** and **MYELOPEROXIDASE**, generates bactericidal hypochlorite
 - also have **DEFENSINS** which act like antibodies
- **Specific granules = secondary granules→**
 - smaller but greater in # than primary granules
 - contain type 4 collagenase, phospholipase, complement activators, and lysozymes
- **Tertiary granules**
 - Contain phosphatases and metalloproteinases which aid in neutrophil migration

Neutrophils

Margination, Rolling (selectins), Adhesion (integrins), Transmigration, Opsonization, Phagocytosis

- Neutrophil migration controlled by expression of adhesion molecules on neutrophil surface - interact with ligands on endothelial cells
- **Selectins** on neutrophil interact with receptors on endothelial cells
- Neutrophil partially tethered to endothelial cell, slowing down and rolling on surface of endothelium

Diagrammatic representation of neutrophil margination, adhesion, rolling, diapedesis, migration into tissues and phagocytosis



Neutrophils (continued)

- Then **integrins** activated by chemokine signals from endothelial cells
- VCAM-1 expressed on neutrophil surface acts with receptors on endothelial cells -- attaches neutrophil to endothelium (adheres)
- Neutrophil extends pseudopod to intercellular junction
- Histamine and heparin released at injury site by mast cells open intercellular junction and allow transmigration of neutrophil into connective tissue

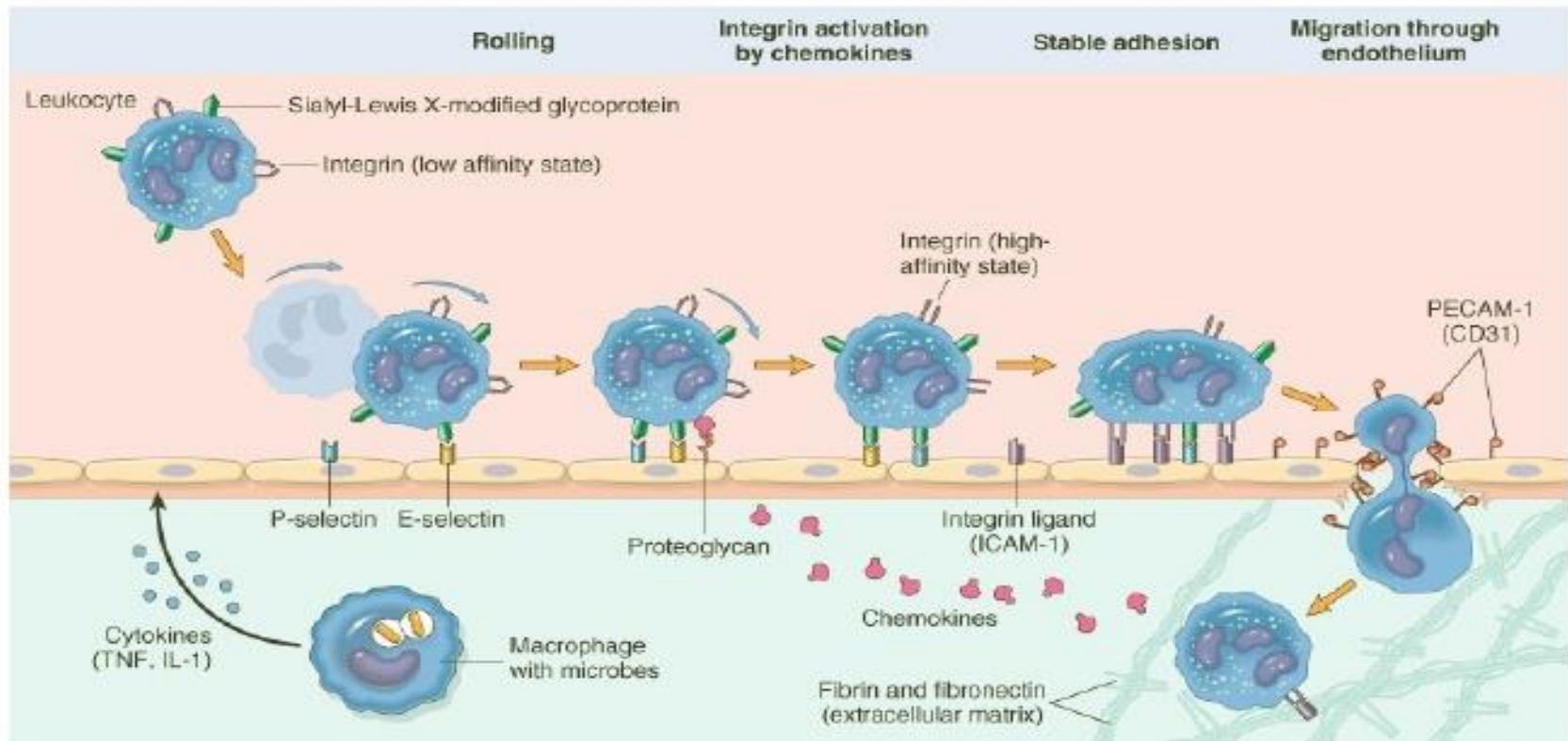
Neutrophils

- Once neutrophil enters connective tissue, further migration to injury site directed by chemotaxis
- Neutrophils need to recognize foreign material before phagocytosis
- Some must **be opsonized** (coated with Abs or complement) to make them more attractive to the neutrophil

Neutrophils

- After recognition and attachment, Ag is engulfed by neutrophil pseudopods - forms phagosome
- Specific and azurophilic granules fuse with phagosome membrane and lysosomal hydrolases of azurophilic granules digest the foreign material
- Neutrophils also secrete IL-1 (pyrogen), induces fever via prostaglandin synthesis acting on thermoregulatory center of hypothalamus

The multistep process of leukocyte migration through blood vessels.



- Slowly move along the endothelium and adhere transiently (process called rolling) finally they come to rest at some point - adhere firmly.
- After adhesion they insert their pseudopods into endothelial cell junction and squeeze through this layer into the extravascular space.
- The process of adhesion and transmigration is determined by binding of adhesion molecules on leukocytes and endothelial cells.

Case

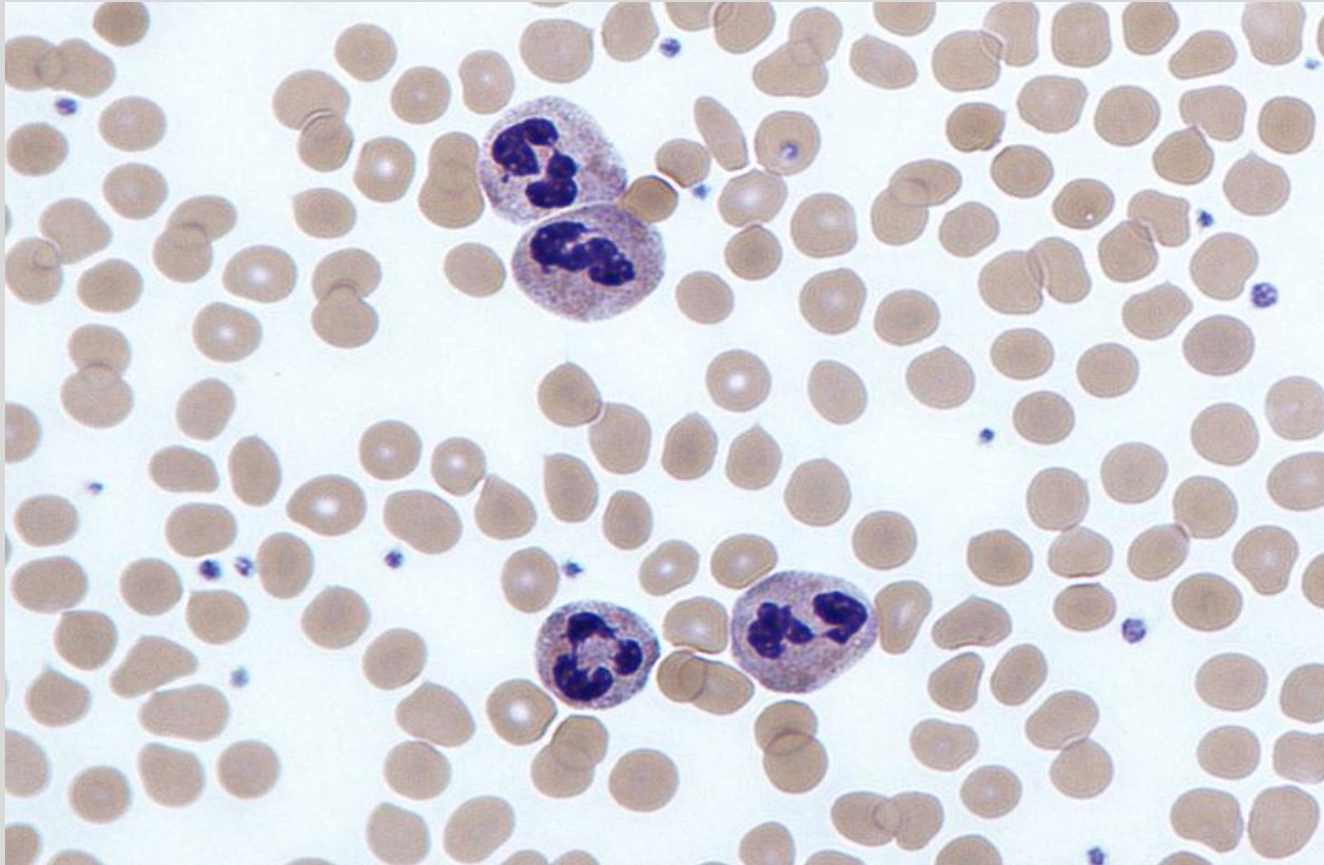
- A 23 year old woman presents to the ED with abdominal pain of 48 hours duration. The pain is sharp, rated 9/10 and localized to the RLQ. She took advil 300 mg with no significant relief. No significant past PMHx, no prior hospitalizations, no surgical hx, no allergies, no Etoh, no drugs, no pertinent family hx.

ROS: fever, nausea, HA, pallor, constipation, dysuria

PE: Abdomen: rigid, tender, + rebound tenderness +

Psoas sign

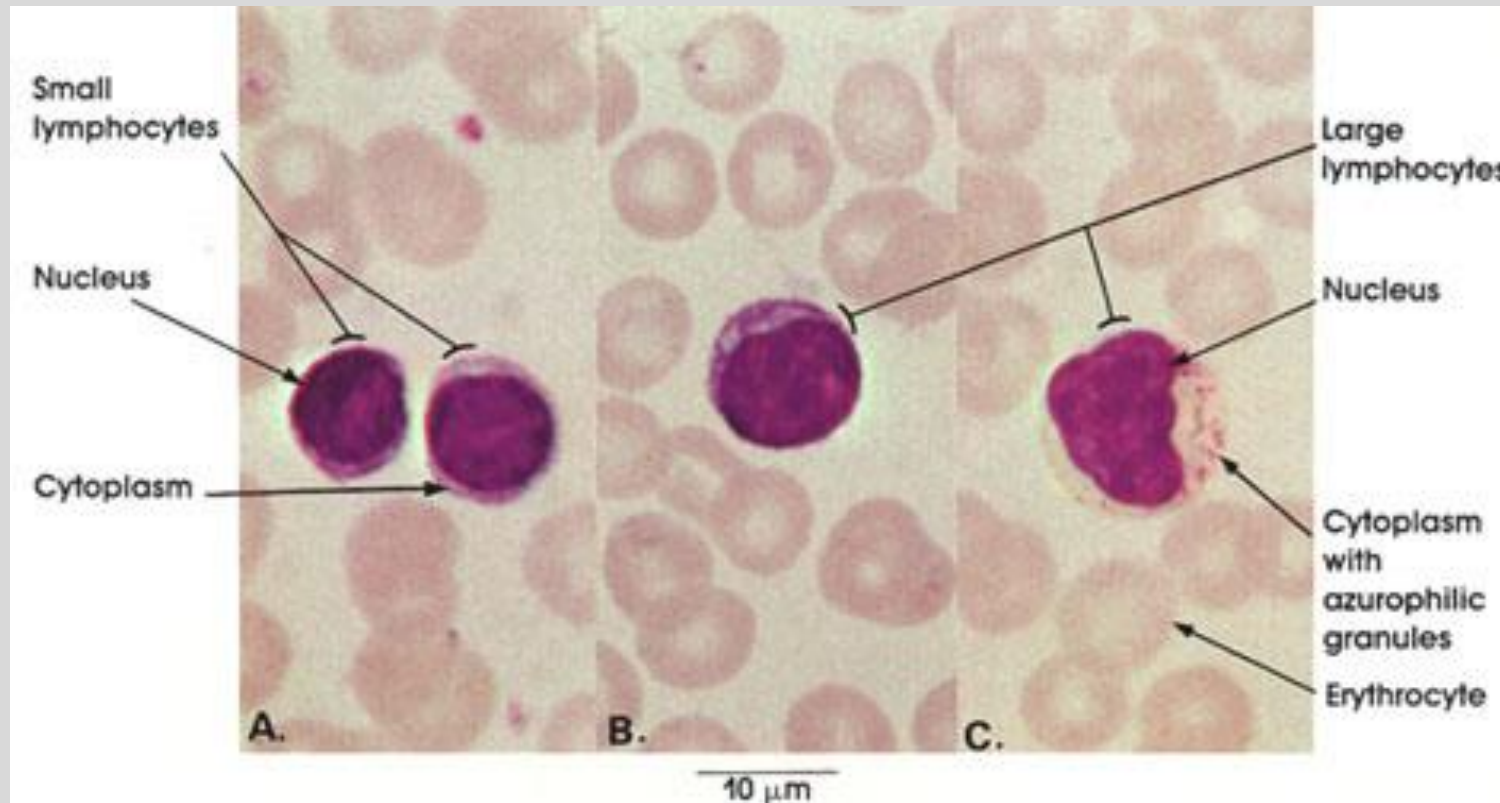
CBC and smear shows:



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LYMPHOCYTES – THINK CHRONIC INFLAMMATION, VIRAL INFLAMMATORY ASSOCIATION



- Most common agranulocytes
- About 30% of total wbc's in peripheral blood
- May see few small, dense lysosomes (azurophilic granules)

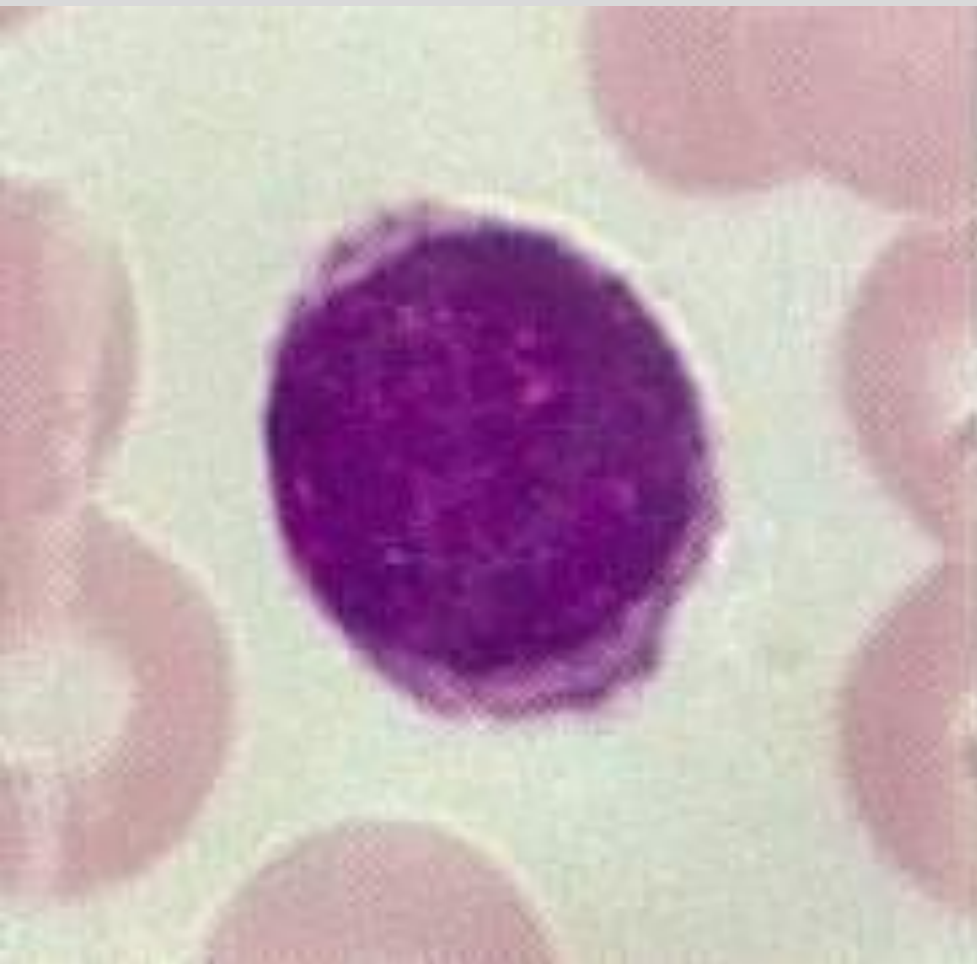
Lymphocytes - Types

- **T lymphs** undergo differentiation in thymus
 - involved in cell mediated immunity. CD 2,3,5,7. Subclassified on basis of CD4+ or CD8+
- **B lymphs** are involved in production of circulating antibodies: CD 19, 20, 24
- **NK lymphs** programmed to kill types of transformed cells (virus infected and some types of tumor cells)
 - usually larger than T and B lymphs, have a kidney shaped nucleus – aka large granular lymphocytes
- Peripheral blood lymphocytes 60 to 70% T lymphs and 20 to 30% B lymphs

CLUSTER OF DIFFERENTIATION

- **CD – cluster of differentiation are Cell membrane molecules that are used to classify leukocytes (wbcs) into subsets**
- CD molecules are defined or classified by the reference monoclonal antibodies to which they bind
- There are many clusters of differentiation; many cells express more than one CD marker.
- In terms of physiology, CD molecules can act in many ways, often acting as receptors or ligands important to the cell.

<https://medical-dictionary.thefreedictionary.com/cluster+of+differentiation>



T or B lymphocyte?

CD proteins

B cells – immunoglobulins

T cells – TCR (T cell receptor)

3 types of T cells

CD – cluster of differentiation - a protocol used for the identification and investigation of cell surface molecules providing targets for immunophenotyping of cells.

Immunohistochemistry via flow cytometry is the only way to differentiate a B cell and a T cell!! Cells are identified via cellular differentiation markers

B cells: CD 19, CD 20, CD 21

T cells: CD4, CD8

NK cell: CD16, CD 56

Cytotoxic CD8+ T lymphocytes:

- primary effector cells in cell mediated immunity
- specifically sensitize and recognize Ags via TCRs on host cells infected by viruses or that have become neoplastic
- only recognize Ags bound to MHC I molecules
- after TCR binds to Ag-**MHC I**, cytotoxic CD8+ cells secrete lymphokines and perforins that produce ion channels in membranes of infected or neoplastic cells, leading to lysis

Major histocompatibility complex (MHC) is a set of cell surface proteins essential for the acquired immune system to recognize foreign molecules

Suppressor CD8+ (CD45RA+) T lymphocytes

- diminish or suppress Ab formation by B cells
- downregulate ability of T lymphs to initiate a cellular immune response — **PLEASE SEE HANDOUT FOR NOMENCLATURE CLARIFICATION**

Helper CD4+ T lymphocytes:

- Needed for induction of immune response to foreign antigens
- Ag bound to **MHC II** molecules presented by Ag-presenting cell (e.g., macrophage) to CD4+ T cell
- Binding of TCR to Ag-MHC II complex activates helper CD4+ cell which then produces mainly IL-2 to stimulate proliferation and differentiation of more CD4+ cells
- These make and secrete lymphokines that affect function and differentiation of B, T, and NK cells
- B cells become plasma cells and make antibody

Monocyte



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Monocyte

- Largest wbc's in peripheral blood; indented, “horseshoe” shaped nucleus, small azurophilic granules (lysosomes)
- During inflammation, monocytes leave blood vessel at site of inflammation, transform into tissue macrophage (secretes IL-1 and TNF-alpha) and phagocytose
- Monocyte-macrophage is an antigen presenting cell
- Life span 3 days
- **In tissue monocytes known as macrophages aka histiocytes**

Macrophages

When monocytes migrate into the designated tissues they assume another name.

macrophages in skin = Langerhans cell

macrophages in CNS = microglia

macrophages in bone = osteoclasts

macrophages in lungs = alveolar macrophages

macrophages in liver = Kupffer cells

Eosinophil



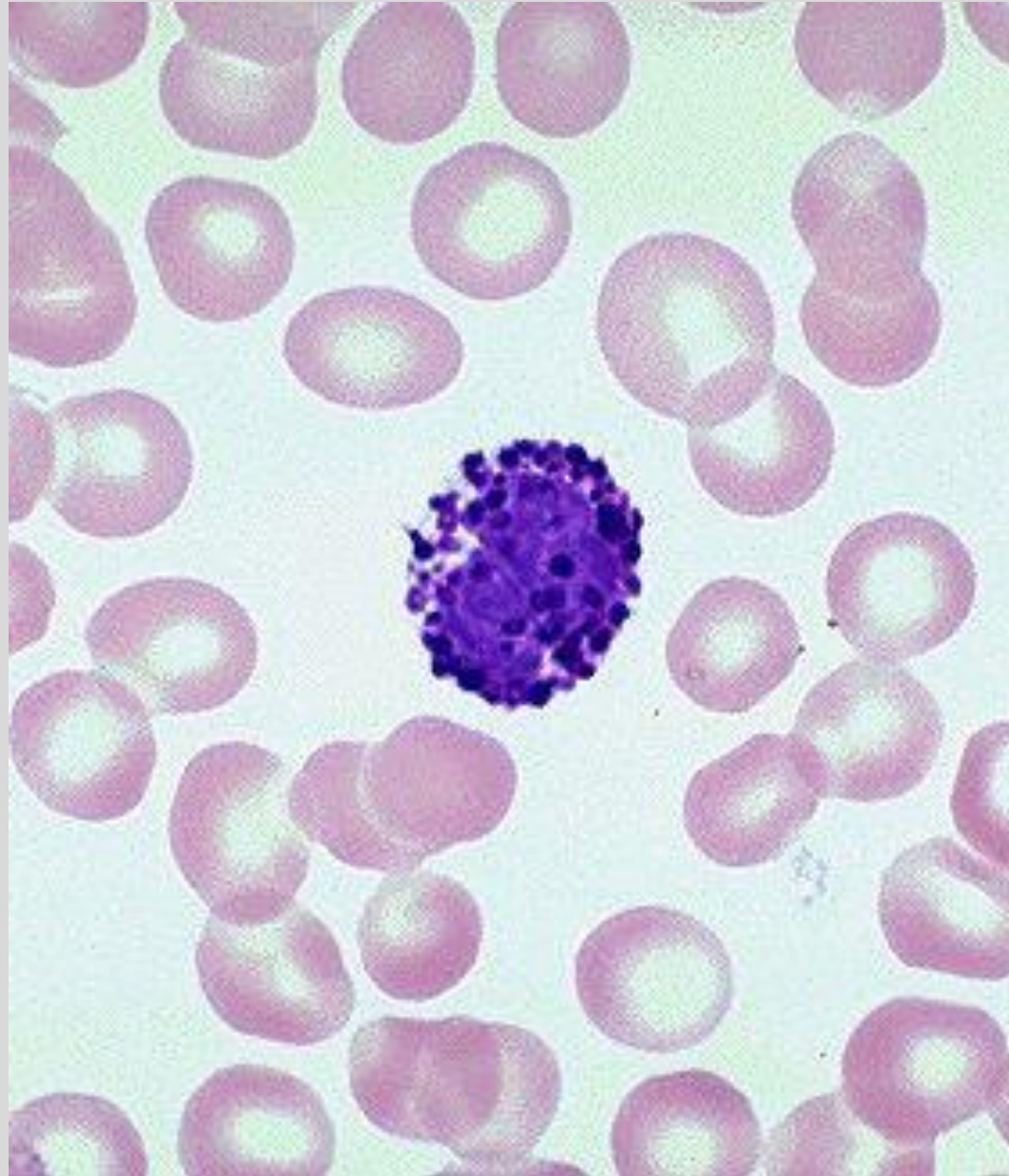
Participate in chronic inflammation, anti-parasitic activity, and allergic reactions

Large, eosinophilic granules; bilobed nucleus

Eosinophils

- Two types of granules:
 - Primary Granules-
 - Azurophilic granules are lysosomes - have enzymes that destroy parasites and hydrolyze Ag-Ab complexes
 - Specific Granules
 - Have a CRYSTALLOID body - contains major basic protein; outer matrix of crystalloid body has eosinophilic cationic protein, eosinophil peroxidase, and eosinophil derived neurotoxin
 - **MBP, EPO,EDN are cytotoxic and destroy parasites. EDN causes NS dysfunction in parasites**
 - Also have histaminase, collagenase, cathepsins, and arylsulfatase which neutralize histamine and leukotrienes secreted by basophils

BASOPHIL



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Basophil

- Least numerous of wbcs
- Plasma membrane has many Fc receptors specific for IgE
- Specific granules have heparin, histamine, heparin sulfate, leukotrienes
- Intense basophilia of the granules because of high concentration of sulfates
- Azurophilic granules are lysosomes
- Functionally related to mast cells – bind IgE via Fc receptors which releases vasoactive agents from basophil granules (causes severe vascular disturbances associated with hypersensitivity and anaphylaxis)



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Platelets

- Aka thrombocytes – small, membrane bounded, **anucleate** cytoplasmic fragments derived from megakaryocytes
- Platelet formation: small bits of cytoplasm are separated from the peripheral regions of the megakaryocyte by platelet demarcation channels
- Life span is about 10 days
- Normal count about 150,000 – 400,000/ul (as with other lab values will vary somewhat from lab to lab; also sometimes different units used)

Platelets

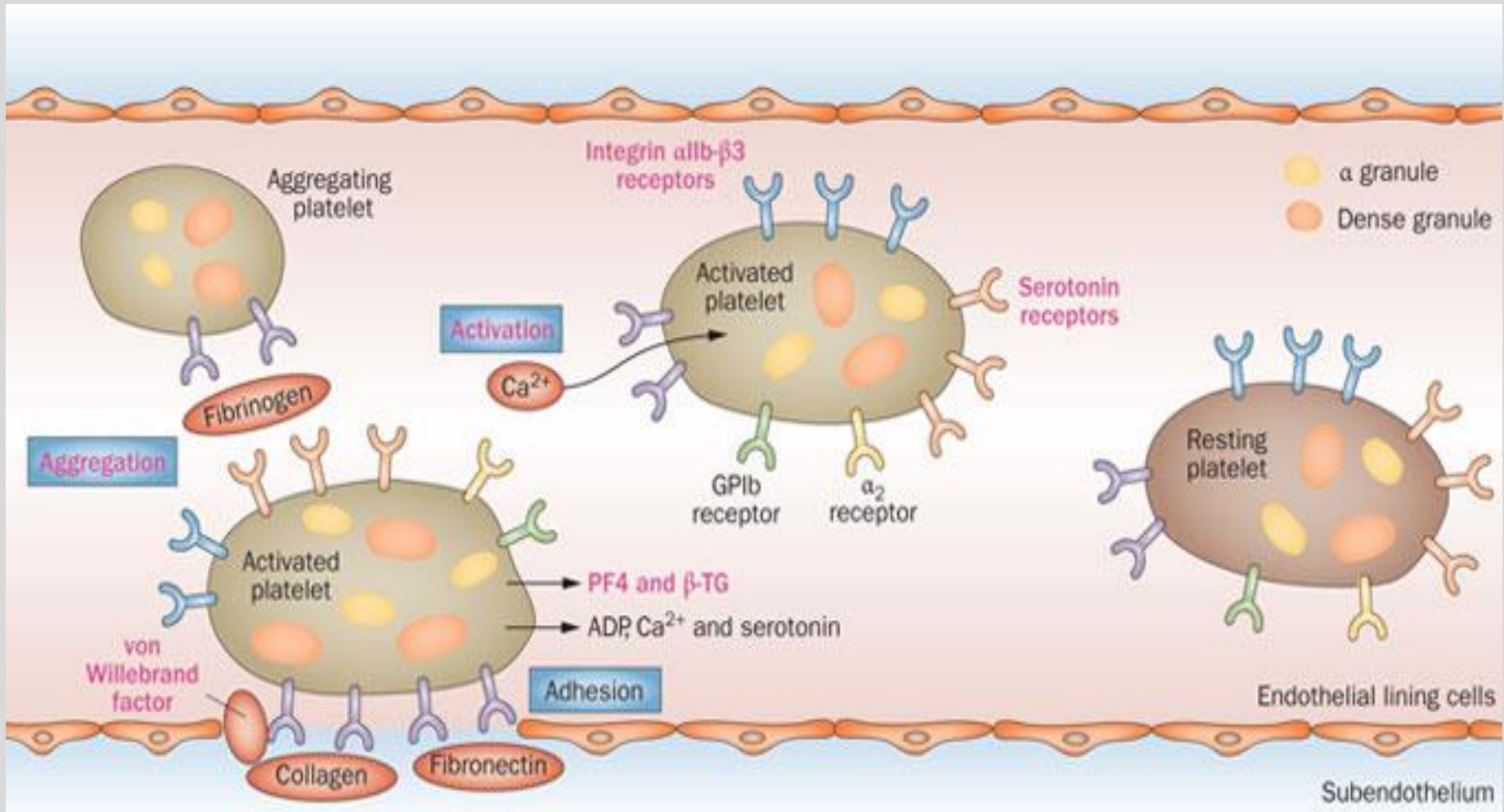
- Alpha granules
- Most numerous
- Contain PDGF, fibrinogen, coagulation factors, plasminogen, plasminogen activator inhibitor
- Important in **vessel repair, blood clotting and platelet aggregation**
- Delta granules
 - Serotonin, ADP, ATP, histamine, serotonin, which help platelet adhesion and vasoconstriction in area of injured vessel
- Gamma/lambda granules
 - Similar to lysosomes and contain hydrolytic enzymes which function in clot resorption during later stages of vessel repair

Platelets

- Platelets: hemostasis –
 - when vessel wall is injured, exposed connective tissue at damage site promotes platelet adhesion
 - Platelets degranulate and release serotonin, ADP, and thromboxane A2
- Serotonin is potent vasoconstrictor, reduces blood flow at injury site
- ADP and thromboxane A2 cause further platelet aggregation into primary hemostatic platelet plug
- Activated platelets also release alpha and delta granules with clotting factors and more serotonin
- Eventual formation of secondary hemostatic plug (via coagulation cascade)

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Case Question

- A 10 year old male was diagnosed with an intestinal worm infection. A blood sample was taken and examined under the microscope. What cell type would predominate?

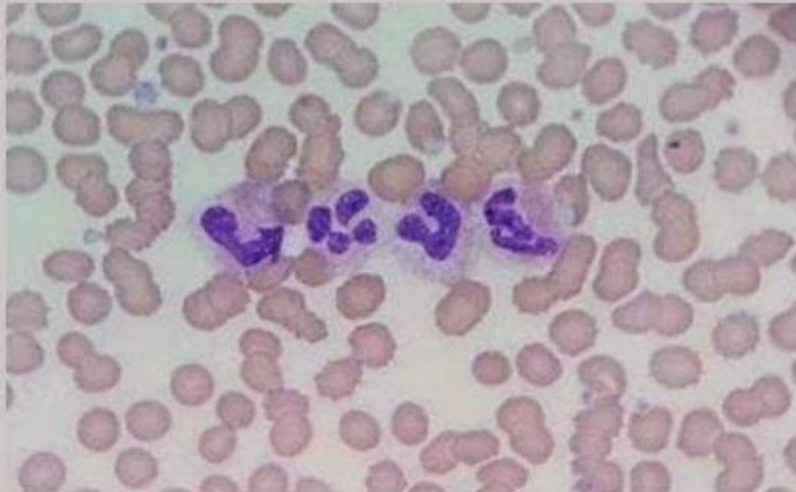
Matching question

- 1. Neutrophil
- 2. Basophil
- 3. Eosinophil
- 4. Lymphocyte
- 5. Monocyte
- 6. Red blood cell
- 7. Platelet
- 8. Plasma
- A. Hemostasis
- B. Contains albumin
- C. Acute bacterial infection
- D. Macrophage
- E. Oxygen delivery
- F. Histamine
- G. Parasites, allergies, asthma
- H. T, B, or NK cells

Answers

- 1. C
- 2. F
- 3. G
- 4. H
- 5. D
- 6. E
- 7. A
- 8. B

LOVE



LOVE IS IN OUR BLOOD

Summary Slide

- Hematocrit – volume of packed rbcs
- Plasma 90% water, albumin, globulins, fibrinogen, electrolytes
- Serum – plasma without clotting factors
- CBC components – Granulocytes include neutrophils, eosinophils, and basophils. Remaining wbc's are lymphocytes, monocytes. Also rbcs and platelets
- Erythrocytes (rbcs) – oxygen carrying
 - Anucleate
 - Biconcave unique structure
 - Normal adult: HbA = two alpha globulin chains, two beta globulin chains

Summary Slide

- Neutrophils (polymorphonuclear cells) – acute, bacterial
 - Primary, secondary, tertiary granules
 - Migration, rolling (selectins), adhesion (integrins), transmigration, opsonization, phagocytosis
- Lymphocytes – chronic, viral
 - T lymphs (CD4, CD8, etc. subsets) – cell mediated immunity
 - B lymphs – produce antibody
 - NK lymphs – kill transformed cells
 - CD markers
- Monocytes (blood)
 - Macrophages (tissue), different names in different tissues
- Eosinophils – granules, parasites, allergies, asthma
- Basophils – granules, related to mast cells, specific granules with heparin, histamine
- Platelets – hemostasis
 - Small, anucleate, alpha, delta, gamma granules

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>